How to use the command line SSH and SFTP clients

Many Unix environments have the command-line SSH and SFTP client software tools installed. This page is intended as a guide to just the basics of using these command-line tools. They have many more features than what is described here -- but these instructions should get you started.

Where would I use these?

CS account

- If you have a CS account, you have access to at least three machines you can log into:
  - `shell.cs.fsu.edu`
  - `program.cs.fsu.edu`
  - `linprog.cs.fsu.edu`

  These machines all mount your home directory, so there's no need to SFTP between them.
- You might also have an account on `websrv2.cs.fsu.edu`. This would **not** mount your normal CS account directory, so you might need to file transfer between this machine and others

Mac OS X home computer / laptop

- If you have a Macintosh laptop or home computer running OS X, you cannot use the Windows SSH client unless you're running Windows emulation or have booted into Windows
- In this case, you can use the `terminal` application (which takes you into a Unix prompt on your Mac) and run the command line SSH and SFTP programs from there
- This is useful for logging into your CS account remotely, as well as doing file transfers between your Mac and your CS account
- To open the terminal application on a Mac:
  - Go to the spotlight icon in the upper right of your desktop
  - In the search box, type "terminal"
  - A menu choice for the terminal application should appear. Select this

Using command-line SSH

Login format:

```
ssh username@host_name
```

In this format, *username* refers to your user name on the remote account you are logging into, and *host_name* refers to the name of the machine (usually along with doman) that you are logging into

If you leave out the username, the command will assume that you are logging into another machine with the *same* username as the machine you're currently on.

Examples

- Suppose I'm on a Mac, logged in with user account "john".
  This command would log me into `shell.cs.fsu.edu` as username "smith":

  ```
  ssh smith@shell.cs.fsu.edu
  ```

  This command would log me into `shell.cs.fsu.edu`, attempting username "john" (probably will fail, since `john` is probably not your CS account username!):

  ```
  ssh shell.cs.fsu.edu
  ```

- You can also log into the machines `program` or `linprog`:

```
ssh jonesb@program.cs.fsu.edu          // username jonesb
ssh jonesb@linprog.cs.fsu.edu
```

Using Command-Line SFTP

SFTP is Secure File Transfer Protocol. It is similar to ssh, but its primary purpose is to enable file transfers between a local machine and a remote machine, whereas the ssh (Secure Shell) protocol is for opening up a general command shell on a remote machine where you have an account.

The login format for command line sftp is exactly the same as with the ssh command, but with the "sftp" command:

```
sftp username@host_name
```

Using the same basis of the prior example, this command would log me into `shell.cs.fsu.edu` with the sftp (file transfer) program as username "smith":

```
sftp smith@shell.cs.fsu.edu
```

Local vs. Remote machines

Before you transfer files, make sure you know the difference between the **local** and **remote** machines:

- The **local** machine is the one you started from -- i.e. the machine you were on when you typed the "sftp" command. This might be your actual physical computer, like a MacBook, if you're using the terminal application. This might also be a machine you are logged into through ssh
- The **remote** machine is the one you just logged into via the sftp command

Examples

- Suppose I'm on my MacBook, with username `john`. I open the terminal application and sftp into `linprog.cs.fsu.edu` with username `smith`:

  ```
  sftp smith@linprog.cs.fsu.edu
  ```

  In this case, the *local* machine is my MacBook, and the *remote* machine is `linprog`

- Suppose I've already logged into user account `smith` on `shell.cs.fsu.edu` through an ssh command (or any SSH client). From my CS account, I type this command:

  ```
  sftp websrv2.cs.fsu.edu
  ```

  Now I have logged into the machine `websrv2`, with the same user account `smith`. In this case, the *local* machine is `shell.cs.fsu.edu`, and the *remote* machine is `websrv2.cs.fsu.edu`.

Basic file transfer commands

There are more commands available than this, but the primary commands you will need are:

- `put` -- copy a file from the local machine to the remote machine
- `get` -- copy a file from the remote machine to the local machine
- `ls` -- get a directory listing on the remote machine
- `cd` -- change your current working directory on the remote machine
- `lls` -- get a directory listing on the local machine
- `lcd` -- change your current working directory on the local machine

Note here that the `ls` (list files) and `cd` (change directory) commands work exactly as you are used to them from a regular unix shell. Except when you use them as-is, you are requesting a listing or a change directory operation on the **remote** machine -- i.e. the machine you just sftp-ed into.

If you want to get a directory listing or change directories on the **local** machine, use the `lls` (local list files) and `lcd` (local change directory) commands instead.

I'm not going to describe these commands (listing and change directories) any further, as readers of this file should already be familiar with these unix commands

`put` and `get` commands

Once you have used the listing and navigation commands to put yourself in the correct directories, use `get` to retrieve files from the remote machine to the local account, and use `put` to send files from your local account to the remote account. The formats of these commands are simple, and you may use unix wildcards.

Format:

```
get remote_path [local-path]
put local_path [remote-path]
```

Note that the second parameter of each command is optional -- and it will serve to specify a different name or destination directory for the transferred file. Most of the time, it's easiest to simply use `put` and `get` with a single parameter, and the destination will default to the current remote or local directory, respectively.

Examples

- This command will copy the file `mycode.cpp` from the remote machine to the local one:

  ```
  get mycode.cpp
  ```

- This command will copy the file `mypaper.doc` from the local machine to the remote one:

  ```
  put mypaper.doc
  ```

- This command will copy the file `assign1.cpp` from the remote machine, and save it as `smith1.cpp` on the local machine:

  ```
  get assign1.cpp smith1.cpp
  ```

- This command will copy all files ending in `.cpp` in the current directory on the local machine to the remote machine:

  ```
  put *.cpp
  ```

Other SFTP commands

To see a full list of SFTP commands and their formats, you can type `help` when you are logged in via `sftp`, and it will give you a list of available commands.

To exit the sftp login, you can use either the `quit` or the `exit` command.